

COMP661 Deep Learning

Semester 2 2026

Assignment 1 - Deep Learning for Soil Organic Carbon Prediction

Worth : 20%

Submission : URL for the GitHub Repository

Dataset : nz_lucas_smap_niwa_dataset.csv

Final Submission Due: Friday, 14 August 2026 5:00 p.m.

Late Penalty: Work submitted after the due date and time attracts a maximum late penalty, as noted below, of the marks available:

Less than 6 hours late	5% late penalty
Between 6 and 24 hours late (i.e., up to 1 day)	12.5% late penalty
Between 24 and 48 hours late (i.e., 1 - 2 days)	25% late penalty
Over 48 hours late (i.e., 2 days or longer)	100% late penalty

No work will be accepted after Sunday, 16 August 2026 5:00 p.m.

This is an **individual** assessment. You must not collaborate or confer with others. You may help others by verbally explaining concepts and making suggestions in general terms, but without directly showing or sharing your own work. You must develop your code, visualisations, and interpretations independently on your own, even if working alongside others. Any work that is copied or contains identical code/visualisations/interpretations to others will receive zero marks for both parties involved.

The use of Artificial Intelligence (AI) tools, such as ChatGPT or Microsoft Copilot, to complete this assessment is **prohibited**. Assessment answers will be analysed for evidence of the use of AI and penalties may be administered.

The University policy on Academic Integrity can be found [here](#).

Introduction:

Artificial Intelligence (AI) and Deep Learning are increasingly used to address environmental and agricultural challenges by modelling complex relationships within large environmental datasets. These models provide efficient alternatives to costly and time-consuming field surveys, supporting better decision-making in environmental monitoring and sustainable land management.

Soil Organic Carbon (SOC) is a key indicator of soil health and agricultural productivity. Accurate prediction of SOC helps monitor soil quality, support sustainable farming practices, and assess land degradation.

In this assessment, you will explore the provided dataset, perform data preprocessing, and develop, optimise, and evaluate an Artificial Neural Network (ANN) model to predict Soil Organic Carbon. You are expected to justify your modelling decisions and critically discuss the performance of your final model.

Dataset Overview

The given dataset integrates environmental observations collected from monitoring locations across New Zealand, sampled under a range of climatic and environmental conditions.

Each observation contains measurements describing:

- geographic location,
- terrain,
- climate,
- soil physical properties,
- soil chemistry,
- vegetation, and,
- environmental quality

The dataset contains

- **5,000 observations**
- **35 predictor variables**
- **1 regression target**

You can find the data dictionary in Appendix A.

Instructions:

Setting up your repository

Create a **private** GitHub repository called **StudentID_EcoVision_Project1** (replace "StudentID" with your actual student ID).

Follow the folder structure given below.

```
123456789_EcoVision_Project1 /
├ README.md
├ requirements.txt
├ data/
│   └── (your raw/provided dataset files go here)
├ src/
│   ├── EDA.ipynb
│   └── DL.ipynb
├ figures/
│   └── (saved visualisations, e.g. target_distribution.png)
└ models/
    └── final trained model files
```

Commit History

Your repository must show at least two meaningful commits per week, from the day you create the repository right through to the final submission deadline. This standard applies across the entire project. Your commit history forms part of the assessment.

Notebook Structure

The project is built around two Jupyter notebooks, both located in the `src/` folder.

1. **EDA.ipynb** contains the exploratory data analysis and preprocessing (answers for Part 1).
2. **DL.ipynb** contains the modelling work and discussion (answers for Part 2 and 3).

The dataset should be loaded using relative paths from within `src/`. Both notebooks must execute successfully from start to finish in a clean environment, using only the packages listed in **requirements.txt**.

Saving Outputs

All key visualisations should be saved to the `figures/` folder using relative paths, so they display directly on GitHub without needing the notebooks to be run. The final trained model should be saved to the `models/` folder. Both folders must be committed to the repository.

Documentation within Notebooks

Structure each notebook with clear Markdown headings. Every code block should be preceded by a Markdown cell briefly outlining what the code does and followed by a Markdown cell interpreting the results it produces. Any preprocessing or modelling decision should be justified with reference to patterns or evidence observed in the data.

All code must be clearly commented throughout.

Visualisation Standards

Every visualisation must include an informative title, labelled axes (with units where relevant), a legend where appropriate, and a short-written interpretation explaining what it reveals about the data. Files saved to the figures/ and models/ folders must use clear, descriptive filenames that indicate their contents at a glance.

Part 1: Data Understanding, Exploratory Data Analysis (EDA), Data Quality Assessment and Data Preprocessing (30 Marks)

Deep learning models are highly dependent on the quality of the input data. Before developing Artificial Neural Networks (ANNs), it is essential to understand the dataset, identify potential data quality issues, explore relationships between variables, and prepare the data appropriately for model training.

In this section, you should develop an initial understanding of the dataset before performing any preprocessing. This involves conducting a comprehensive exploratory data analysis (EDA), assessing the quality of the dataset, applying suitable preprocessing techniques, and investigating the importance of predictor variables.

For each key finding in your EDA (e.g., distributions, relationships between variables, outliers, missing values, class imbalance), briefly state what you observed and why it matters.

For each preprocessing step applied (e.g., handling missing values, treating outliers, encoding categorical variables, scaling, feature selection, train/test split), briefly state what you did and why.

Your work should be presented in a well-structured Jupyter Notebook (**EDA.ipynb**) with clear explanations, tables, and visualisations.

Part 2 – Model Development and Improvements (50 Marks)

Following the completion of data understanding and preprocessing, the next stage is to develop baseline Artificial Neural Network (ANN) model. A baseline model provides an initial benchmark against which improved architectures and optimisation strategies can be evaluated.

Task 2.1 Initial model development

Develop a baseline ANN to predict Soil Organic Carbon (**soil_organic_carbon_target**) using a single hidden layer with 32 neurons with ReLU activation function.

Compile the model using:

- **Loss Function:** Mean Squared Error (MSE)
- **Optimizer:** Adam (default learning rate)
- **Evaluation Metric:** Mean Absolute Error (MAE)

Train the model using:

- **Epochs:** 100
- **Batch size:** 32
- **Validation set:** created in Part 1

Set a random seed prior to training to ensure your results are reproducible.

Task 2.2 – Model Improvements

Conduct a systematic investigation to optimise the deep neural network architecture by completing the following tasks:

- Evaluate the effect of network depth by developing models with 1 - 4 hidden layers.
- Using the most suitable depth identified, investigate the effect of varying the number of neurons per hidden layers.
- Using the best architecture so far, evaluate the effect of different activation functions.
- Investigate the effect of different optimisation algorithms, including appropriate tuning of their associated hyperparameters.
- Evaluate suitable regularisation and generalisation techniques to reduce overfitting and improve model performance.

Summarise and compare the results of all experiments in well-organised tables and provide a justification for the final model selection based on the experimental findings.

Task 2.3 – Selection of the Final Model

Using the findings in Tasks 2.2, develop your final regression model.

Present the complete model specification including:

- Number of hidden layers
- Number of neurons in each layer
- Activation functions
- Optimiser
- Learning rate
- Batch size and number of epochs (or early stopping criterion, if used)
- Regularisation techniques

Support your architecture choices with evidence from your Task 2.2 experimental results, explaining why this configuration was selected over the alternatives tested.

Part 3 – Model Interpretation, Critical Evaluation and Professional Reporting (20 Marks)

In this final section, you will evaluate the best-performing regression model developed in Task 2.3, and critically discuss its performance and limitations. All conclusions should be supported by evidence gathered throughout the assignment.

Task 3.1 – Model Interpretation and Critical Evaluation

1. Model Comparison

Summarise the performance of the baseline and final models in Table 1, using your test set results, reported to an appropriate number of decimal places.

Table 1: Model comparison

Model	MAE	MSE	RMSE	R ²
Baseline				
Final				

2. Performance Discussion

Using the results in Table 1 and any supporting plots (e.g., predicted vs. actual plot), discuss:

- How accurately does the final model predict Soil Organic Carbon?
- Has prediction accuracy improved compared with the baseline model?
- Are prediction errors randomly distributed, or is there a pattern?
- Are there specific observations or ranges that are consistently difficult to predict?

3. Critical Evaluation

Critically evaluate the strengths and limitations of your final deep learning model. Your discussion should address:

- The contribution of key hyperparameters to model performance.
- How overfitting was identified and addressed during model development.

Task 3.2 – Professional Report and GitHub Repository

Jupyter Notebooks

Your notebooks should:

- Run from beginning to end without errors.
- Contain clear section headings.
- Include Markdown explanations before major code sections.
- Present figures and tables with appropriate titles and captions.
- Provide concise interpretations of results.

README.md

Include:

- Project title
- Student name and ID
- Brief project overview
- Dataset description
- Installation instructions
- Required Python packages
- Instructions for running the notebook
- Summary of key findings

Code Quality

Your notebook should demonstrate:

- Appropriate code organisation
- Meaningful variable names
- Clear comments where necessary
- Consistent formatting
- Reproducible results (e.g., fixed random seeds)

Marking guide is included in the Appendix B.

Appendix A – Data Dictionary

Dataset Name: nz_lucas_smap_niwa_dataset.csv

Number of Records: 5,000

Predictor Variables: 36

Target Variables: 1

Table 2: Geographic Variables

Variable	Description	Unit	Type	Typical Range	Source
latitude	Latitude of the monitoring location	Decimal degrees (°)	Numeric	-47 to -34	LUCAS
longitude	Longitude of the monitoring location	Decimal degrees (°)	Numeric	166 to 179	LUCAS

Table 3: Terrain Variables

Variable	Description	Unit	Type	Typical Range	Source
elevation_m	Elevation above mean sea level	metres (m)	Numeric	0–2200	DEM
slope_deg	Average terrain slope	degrees (°)	Numeric	0–45	DEM
aspect_deg	Direction the slope faces	degrees (0–360°)	Numeric	0–360	DEM
topographic_wetness_index	Index representing water accumulation potential	Index	Numeric	1–15	DEM
terrain_ruggedness_index	Terrain roughness index	Index	Numeric	0–15	DEM

Table 4: Climate Variables

Variable	Description	Unit	Type	Typical Range	Source
annual_rainfall_mm	Mean annual rainfall	mm/year	Numeric	400–3500	NIWA
annual_temperature_c	Mean annual air temperature	°C	Numeric	3–20	NIWA
summer_temperature_c	Mean summer temperature	°C	Numeric	8–28	NIWA
winter_temperature_c	Mean winter temperature	°C	Numeric	-2–15	NIWA
solar_radiation_mj	Average daily solar radiation	MJ/m ² /day	Numeric	8–22	NIWA
potential_evapotranspiration_mm	Annual potential evapotranspiration	mm/year	Numeric	500–1400	NIWA
relative_humidity_pct	Average annual relative humidity	%	Numeric	40–100	NIWA

frost_days_per_year	Average annual frost days	days/year	Numeric	0–120	NIWA
drought_index	Annual drought severity index	Index	Numeric	0–5	NIWA

Table 5: Soil Physical Properties

Variable	Description	Unit	Type	Typical Range	Source
clay_pct	Clay content	%	Numeric	5–60	LUCAS
sand_pct	Sand content	%	Numeric	10–85	LUCAS
silt_pct	Silt content	%	Numeric	5–70	LUCAS
bulk_density	Dry bulk density	g/cm ³	Numeric	0.8–1.8	LUCAS
soil_depth_cm	Effective soil depth	cm	Numeric	20–200	LUCAS
coarse_fragment_pct	Coarse fragments (>2 mm)	%	Numeric	0–50	LUCAS
available_water_capacity_mm	Plant available water capacity	mm	Numeric	10–180	Derived

Table 6: Soil Chemical Properties

Variable	Description	Unit	Type	Typical Range	Source
soil_ph	Soil acidity/alkalinity	pH	Numeric	4.5–8.5	LUCAS
organic_nitrogen_pct	Organic nitrogen concentration	%	Numeric	0.1–3.5	LUCAS
available_phosphorus_mgkg	Available phosphorus	mg/kg	Numeric	2–80	LUCAS
exchangeable_potassium_mgkg	Exchangeable potassium	mg/kg	Numeric	20–500	LUCAS
cation_exchange_capacity	Cation exchange capacity	cmol(+)/kg	Numeric	3–45	LUCAS

Table 7: Remote Sensing Variables

Variable	Description	Unit	Type	Typical Range	Source
ndvi_mean	Normalised Difference Vegetation Index	Unitless	Numeric	0.05–0.95	SMAP/Satellite
evi_mean	Enhanced Vegetation Index	Unitless	Numeric	0.02–0.90	SMAP/Satellite

ndwi_mean	Normalised Difference Water Index	Unitless	Numeric	-0.30–0.70	SMAP/Satellite
soil_moisture_pct	Average volumetric soil moisture	%	Numeric	5–80	SMAP

Table 8: Environmental Quality Indicators

Variable	Description	Unit	Type	Typical Range	Source
soil_quality_index	Composite soil quality score	Index (0–100)	Numeric	0–100	Derived
climate_stress_index	Composite climate stress score	Index (0–100)	Numeric	0–100	Derived
erosion_risk_index	Estimated soil erosion risk	Index (0–100)	Numeric	0–100	Derived

Target Variable

Soil Organic Carbon Prediction

Table 9: Target Variable

Variable	Description	Unit	Type	Typical Range
soil_organic_carbon_target	Soil Organic Carbon content	tonnes of carbon per hectare (t C/ha)	Continuous	5–120

Purpose: Predict the amount of carbon stored in the soil using environmental, climate, terrain, vegetation, and soil properties.

Appendix B

Table 10: Marking Guide

Section	Description	Criteria	Marks
1	Dataset Understanding and EDA	Comprehensive exploration using appropriate visualisations and insightful interpretations. Data quality issues identified correctly. Relationships between variables are clearly explained.	15
	Preprocessing	Appropriate preprocessing methods selected and thoroughly justified. Appropriate feature importance methods used. Results interpreted critically and linked to modelling decisions.	15
2	Baseline ANN Development	Correct implementation of baseline architecture, reproducible experiments, and clear explanation of model configuration and results.	5
	Systematic Hyperparameter Investigation	Comprehensive experimental investigation of architecture depth, neurons, activation functions, optimisers, hyperparameters and regularisation. Experiments are systematic, justified and well documented.	40
	Final Model Selection	Final architecture is fully justified using experimental evidence. Architecture, hyperparameters and regularisation techniques are clearly documented.	5
3	Model Evaluation and Critical Discussion	Thorough comparison between baseline and final models using appropriate metrics. Excellent interpretation of results, strengths, limitations and overfitting/generalisation. Conclusions supported by evidence.	10
	Professional Presentation and GitHub Repository	Repository is well organised with required structure, reproducible notebooks, meaningful commit history, clear documentation, excellent code quality and professional presentation.	10
		Total	100